Cobot Path Optimization Considering Obstacles, Vision and Joint Velocity Constraints

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Abstract

This paper addresses the problem of task sequencing and path optimization for collaborative robots in aerospace manufacturing processes. Unlike conventional approaches that minimize travel distance in task space, the proposed method considers joint velocity constraints, obstacle avoidance, and vision-induced motion requirements. A unified cost function integrating joint-space travel time, back-off motion, vision acquisition, and task execution is formulated. The optimal task sequence is obtained by minimizing the total cycle time. Experimental results on a drilling task with 100 target points demonstrate that the proposed method achieves up to 3.1% reduction in total cycle time compared to conventional approaches. The results highlight the importance of joint-space-aware planning in real-world robotic applications.

1. Introduction

In aerospace manufacturing processes, tasks such as drilling, riveting, and sealing require high precision and repeatability. The overall cycle time is highly dependent on task sequencing and robot motion paths. Recently, collaborative robots (cobots) have been widely adopted due to their flexibility and safety, enabling multiple operations using different end effectors on a single platform.

The integration of 3D vision systems further enhances automation by providing real-time information about target positions and environmental conditions. However, conventional path planning methods that minimize travel distance in task space are insufficient to minimize total execution time in real-world environments.

Existing studies mainly focus on minimizing TCP-based travel distance or optimizing task sequences independently. However, real collaborative robot systems involve the following constraints:

(1) Different maximum velocities for each joint
(2) Inaccessible regions due to obstacles and fixtures
(3) Vision constraints requiring back-off motion before sensing
(4) Minimization of total cycle time including motion, sensing, and execution

Since collaborative robots operate under joint velocity constraints, the shortest path in task space does not necessarily correspond to the minimum execution time. Additionally, vision-based operations require mandatory back-off motions, which introduce additional costs.

In this study, we formulate a task sequencing problem that considers joint velocity constraints and vision-induced motions, and propose a method to minimize the total cycle time.

2. Method

2.1 System Configuration

The system consists of a collaborative robot, end effectors, a 3D vision system, and a control computer. Before task execution, the robot moves to an initial position. The vision system detects task points and obstacles and sends this information to the control system.

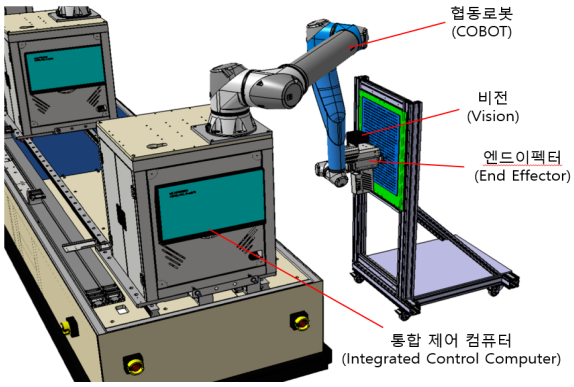

Fig. 1. System Configuration

2.2 Problem Formulation

We define the optimization problem considering both obstacle constraints and vision requirements. Let W denote the workspace, O the obstacle region, and F the free space:

$$W \subset R^3 \tag{1}$$

$$O \subset W \tag{2}$$

$$F = W \setminus O \tag{3}$$

The set of task points is defined as:

$$P = \{p_1, p_2, \ldots, p_N\}, \quad p_i \in F \tag{4}$$

The cost components are defined as follows:

- Travel cost:

$$C_{travel}(i,j) = \max_{k=1,\ldots,6}\left(\frac{|\theta_k^{(j)} - \theta_k^{(i)}|}{\dot{\theta}_k^{\max}}\right) \tag{5}$$

If a collision is expected, $C_{travel}$is set to infinity.

- Back-off cost:

$$C_{back}(i) = \frac{\sqrt{(\triangle y)^2 + (\triangle z)^2}}{v} \tag{6}$$

- Vision cost:

$$C_{vision}(i) = M(t_{vis} + \lambda(1 - P_{suc}(d_i))) \tag{7}$$

- Task execution cost:

$$C_{task}(i) = t_{proc} \tag{8}$$

The total cost for a sequence $\pi$is defined as:

$$J(\pi) = \sum_{k=1}^{N-1} C_{travel}(p_{\pi_k}, p_{\pi_{k+1}}) + \sum_{k=1}^{N}(C_{back}(k) + C_{vision}(k) + C_{task}(k)) \tag{9}$$

The optimal sequence is obtained by:

$$\pi^* = \arg\min_{\pi} J(\pi) \tag{10}$$

2.3 Experimental Results

We evaluated the performance of the proposed method on a drilling task with 100 target points. The following methods were compared:

- Proposed Optimization (PO)
- Nearest Neighbor (NN)
- Spiral (SP)
- Raster (RA)

All experiments were conducted under identical conditions, and execution time was measured in seconds.

Table 1. Test Result

구분	PO	NN	SP	RA
C_{travel}	312	319	323	340
C_{back}	40	40	40	40
C_{vision}	32	32	32	32
C_{task}	500	500	500	500
C_{total}	884 (96.9%)	891 (97.7%)	895 (98.1%)	912 (100%)

The proposed method achieved a total execution time of 884 seconds, corresponding to 96.9% of the raster method (921 seconds), resulting in a 3.1% improvement. NN and SP methods resulted in 891 seconds and 895 seconds, respectively.

The difference in execution time mainly originated from the travel cost, while other costs remained constant. This indicates that joint-space motion significantly impacts total execution time.

These results confirm that minimizing travel distance in task space is not sufficient, and joint-space-aware optimization is essential for real-world robotic applications.

3. Conclusion

This study addressed the task sequencing problem for a 6-DOF collaborative robot considering joint velocity constraints. Conventional methods such as nearest neighbor, spiral, and raster approaches focus on minimizing task-space distance, but fail to account for joint motion constraints.

We proposed a unified time-based cost model that integrates motion, vision, and task execution. The experimental results demonstrated a 3.1% reduction in cycle time compared to conventional methods.

The results highlight the necessity of time-based optimization considering joint constraints in collaborative robot applications such as drilling, riveting, and sealing.

4. Future Work

Future research will extend the model to multi-objective optimization by incorporating joint torque, dynamic load, and fatigue. This will enable not only cycle time reduction but also

improvements in system durability and stability.

References

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